

Name of Student: Husam Nour	PRN:25070122086
Semester: III	Year : AY 2026-27
Subject: Programming with Java	
ASSIGNMENT No : 2	
TITLE : Constructors in Java	

Problem Statement

Develop a Java application demonstrating the use of default, parameterized, and copy constructors for object initialization and data management.

Learning Objectives

To understand and implement default, parameterized, and copy constructors for object initialization.

Programme

1) Default Constructor

CODE

```
class Student{
    String name;
    int age;
    Student(){
        name = "husam";
        age =20;
    }
    void display(){
        System.out.println("Name:" + name + ", Age:" + age);
    }
}
public class demo{
    public static void main(String[] args){
        Student student1 = new Student();
        student1.display();
    }
}
```

Output

```
[Running] cd "c:\Users\husam\OneDrive\Desktop\JAVA_PRACTICE"
Name:husam, Age:20

[Done] exited with code=0 in 0.795 seconds
```

2) Parametrized Constructor

CODE

```
class Student{
    String name;
    int age;

    Student(String n,int a){
        name = n;
        age = a;
    }

    void display(){
        System.out.println("Name:" + name + ", Age:" + age);
    }
}

public class demo{
    public static void main(String[] args){
        Student student1 = new Student("John",20);
        student1.display();
    }
}
```

Output

```
[Running] cd "c:\Users\abell\OneDrive\Desktop\JAVA_PRACTICE"
java demo
Name:John, Age:20

[Done] exited with code=0 in 0.792 seconds
```

3) Copy Constructor

CODE

```
class Student{
    String name;
    int age;

    Student(String n,int a){
        name = n;
        age = a;
    }

    Student(Student s){
        name = s.name;
        age = s.age;
    }

    void display(){
        System.out.println("Name:" + name + ", Age:" + age);
    }
}

public class demo{
    public static void main(String[] args){
        Student student1 = new Student("David",20);
        Student student2 = new Student(student1);

        student1.display();
        student2.display();
    }
}
```

Output

```
[Running] cd "c:\Users\abell\OneDrive\Desktop\JAVA_PRACTICA
Name:David, Age:20
Name:David, Age:20

[Done] exited with code=0 in 0.892 seconds
```

Conclusion

This program demonstrates the use of constructors in Java through the Student class. The default constructor assigns fixed values, the parameterized constructor allows custom student data, and the copy constructor creates a

new object using an existing student's values. Together, they show how object initialization can be handled flexibly in Java.